Irrigation Management in Coffee

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Introduction

Coffee is one the most important plantation crops in India. The major species grown in India are *Coffea arabica* and *coffea canephora*, commonly known as arabica and robusta coffee respectively. Among the two species, robusta is more susceptible for drought and responds well to irrigation than arabica. Coffee is cultivated in various agro-climatic conditions of Western Ghats with varying rainfall conditions of low to very high. Most of the coffee growing areas receive rainfall only during the Southwest monsoon to an extent of 75-80 % of the total annual rainfall. Remaining periods will be dry. An analysis made on rainfall and potential evapotranspiration (1980-2000) at Central Coffee Research Institute has indicated that even though the rainfall is surplus by 1842.58 mm, the water deficit accounts to 248.2 mm during the winter and summer months, especially during the critical periods. Any failure of rains during the blossom leads to partial to complete crop loss both in arabica and robusta coffee. This made coffee to depend more on irrigation to over come uncertain yield and drought situations. Hence, under this article irrigation management is dealt completely.

Healthy blossom in robusta coffee

Pinking of flower buds in robusta

Critical Periods in coffee

The critical periods of coffee growth and development falls usually towards the dry seasons of any year. The most critical periods of coffee are

1. Blossom and backing period
2. Berry expansion stage
3. Dry matter development and bean filling stage

Snakemouth flowers

The blossom and backing periods for both arabica and robusta fall during summer months. The normal blossom shower period for robusta is February-March and for arabica it is March – April. The backing showers are required within three weeks of first blossom shower and the period normally fall between March-April and April to May for robusta and arabica respectively. Normally other two stages do not face much water shortage due to the onset of monsoon by June every year. In case if rains are delayed after backing showers, irrigation can be given to over come these critical periods.

Star flowers

Rains are mostly uncertain during February to April. Some times, these showers may or may not come leading to drying up of flower buds of coffee due to high temperatures, especially in thin shade or open conditions. Among these three critical periods, the most important is the blossom and backing periods; failure of rains during these periods will lead to severe crop damage.

Effects of failure of blossom and backing showers

It is most important to know the ill effects of failure of rains during the blossom and backing periods. Not only the time of showers but also the quantity of showers is very important for healthy blossom and fruit set. A rainfall of 40-50 mm is required for healthy and uniform blossom. If rains are insufficient, blossom will be either partial or incomplete leading to following different floral abnormalities.

Blossom showers - Abnormalities

Star flowers: Most common in arabica. Whenever the rains are very meager or less, only the calyx of flower buds open and green star shaped calyx called starflowers are formed, which do not develop into fruits at all.

Snake mouth flowers: Most common in arabica. When rainfall received is slightly good but not sufficient for full blossom, petals of arabica flower buds move to candle stage and do not open completely. Only bifid stigma of the flowers comes out of buds resembling the mouth of a snake and dry up without fertilization/ pollination. These flowers are termed as snakemouth flowers.

Pinking and paddying: These are the most common floral abnormalities found in robusta due to shortage of water during blossom period. When rains are insufficient, the flower buds move to candle stage and turn pink without opening. This is called pinking. Some times, due to high temperature and shortage of water, these moved buds dry up and turn black called paddying of buds.

In all the above cases, buds will not at all open leading to failure of blossom and ultimately crop loss.

Backing showers - Effects

Once sufficient blossom showers are received, flower buds open by 9-10th day. Coffee requires one more rainfall usually called backing showers within 20-25 days after blossom. This shower is most important for better fruit set and completion of blossom of all the matured buds. If there is any failure in backing showers, there will be considerable reduction in yield due to ovule abortion and improper fruit set. In any case, if rains are received on full blooming day especially in the morning, there will be complete washout of pollen grains leading to non pollination and ultimately crop failure.

Under these circumstances, irrigation plays a major role in overcoming these natural irregularities and to get assured crop from coffee.

Irrigation management

Once irrigation is thought, it is very important to know about the quantity, frequency and methods of irrigation depending on the season, rainfall and soil moisture.

Quantity of irrigation

The quantity of irrigation water required for blossom is 40-50 mm. That means for every one-acre, one and half to two-lakh liters of water. Where as, for backing irrigation water required is 25mm i.e. one-lakh liters of water for every one acre. Irrigation water required for winter and summer period is 25mm per irrigation. Most of the times, the quantity of irrigation water required per unit area is not understood by many planters and approximately irrigate for 5-6 hours using sprinkler without considering the actual discharge of the system. The formula to calculate irrigation water requirement is given below.

Water Required (Liters) = Area (m^2) x Irrigation water (mm)

Blossom irrigation water required for one acre = 4000 m^2 x 40mm = 1,60,000 liters

Backing irrigation water required for one acre = 4000 m^2 x 25mm = 1,00,000 liters

Frequency of irrigation

Frequency of irrigation theoretically should be repeated whenever the available soil moisture falls down to < 50 % of field capacity of the soil. This is usually once in 15-25 days depending on the depletion of soil moisture and soil type. If soils are sandy-to-sandy loam, the frequency may increase i.e. more number of irrigations are required. Otherwise, if soils are silt to clay loam, then the frequency may come down and lesser number of irrigations is required.

Irrigation of Robusta: Robusta can be irrigated for winter @ 25 mm once in 15-25 days beginning from 15-20 days after cessation of normal monsoon and continued till the end of December. Later a gap should be given for a period of 45 days to create moisture stress for getting uniform blossom. Blossom irrigation has to be given during the 2nd fortnight of February followed by a backing irrigation. The backing irrigation has to be given within 20-25 days after blossom. Later if summer showers, robusta can be irrigated @ 25mm once in 15-25 days. Winter and summer irrigation can be practiced only if sufficient irrigation water is available. Else only blossom and backing irrigation can be given.

Table 1. Yield of robusta coffee as affected by sprinkler irrigation (1969-72)

Treatments	CCRI		Bhutankadu Estate	
	Ripe Cherry Kg/ha	% increase	Ripe Cherry Kg/ha	% increase
A – Control	4003	-	5185	-
B – Irrigation for blossom	5391	35	5984	15
C – Irrigation for blossom and backing	6275	57	6577	27
D – Continuous irrigation so that ASM does not fall < 50 % FC	7816	95	8138	47

(source: Awatramani *et al.*, 1973)

Irrigation of Arabica: Being drought tolerant, usually arabica is not irrigated. It can be irrigated either for blossom or backing depending on the rainfall only to substitute the additional water requirement whenever the rainfall is less.

Methods of irrigation

Most common method of irrigation in coffee is sprinkler system due to its many advantages like possibility of shifting irrigation, easiness to irrigate on slope, creation of cool microclimate, requirement of less skilled labour etc. However, while practicing sprinkler irrigation, many planters waste water by excessive irrigation or irrigate less without good knowledge on the system.

Practical hints for sprinkler irrigation

- Instead of zigzag manner always space the jets in square manner so that area of irrigation can be easily calculated
- In case of large holdings, divide the entire area into 20 blocks and start irrigating for blossom from the 1st to last block @ one block per day using required number of shifts. Then immediately begin from the 1st block for backing irrigation so that all the blocks are irrigated equally without any delay in backing irrigation.
- Space the jets so that overlapping of throws should be at least 60 % but not 100 % or more with uniform pressure. After stabilization of pressure, measure discharge rate of jets in liter per minute from different jet points to calculate operating hours for irrigating each area using the following equation.

Irrigation water required (liters) = Area of jets placed (m^2) x Irrigation quantity (mm)

Operating hours = Irrigation water (liter)/(Discharge (liter/min) x 60 x number of jets)

Other methods of irrigation

Other methods that are tested in research stations were drip and micro-sprinkler irrigation. Drip irrigation was found efficient only to maintain young coffee and for sustenance irrigation during the dry period. It could not induce blossom in coffee. Micro-sprinkler proved good for inducing blossom and getting yield equal to sprinkler irrigation by consuming only 50-60 % of overhead sprinkler irrigation water in robusta. However, these methods can be adopted only in small areas but not on large areas due to their some drawbacks like clogging of nozzles, non-shifting irrigation etc.

Research findings of irrigation in coffee

India is probably one of the few countries that have conducted extensive studies on sprinkler irrigation in coffee. Starting from 1959, a series of experiments were carried out by CCRI at different locations for a period of nearly 20 years. Initial experiments were concentrated mostly on arabica coffee, which did not yield any positive responses. Later the experiments on robusta coffee have yielded most significant results, which formed the basis for present day recommendation on robusta irrigation. The studies conducted during 1960s and 1970s indicated a yield increase of 15-95 % in robusta over unirrigated control (Table 1). In another study at CCRI during 1997-2000, a yield increase of 39-110 % was obtained with various methods of irrigation in robusta coffee (Table 2).

Table 2. Yield of clean coffee Kg/ha under various irrigation methods

Treatments	Mean yield (1997-2000)	% Increase over control
T1 – Control	821	-
T2 – Sprinkler for blossom and backing	1327	61.50
T3 – Sprinkler for winter, blossom and backing	1732	110.70
T4 – Microsprinkler for winter, blossom and backing	1434	74.50
T5 – Daily drip irrigation @ 8 LPD	1144	39.30
CD 5 %	351.54	-

(Source: Hariyappa *et al*, 2000.)

Conclusion

As rainfall is erratic and uncertain during the current decades, irrigation management plays major role in coffee, especially in robusta. Timely blossom and backing irrigation not only saves coffee but also increases yields significantly.

References

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